Binomial Theomrem

256. If $\sum_{i=1}^{20}\left(\frac{{}^{20}C_{i-1}}{{}^{20}C_i + {}^{20}C_{i-1}}\right)^3 = \frac{k}{21}$, then k equals :

(A) 400 (B) 100 (C) 200 (D) 50

यदि $\sum_{i=1}^{20}\left(\frac{{}^{20}C_{i-1}}{{}^{20}C_i + {}^{20}C_{i-1}}\right)^3 = \frac{k}{21}$, तो k बराबर है :

(A) 400 (B) 100 (C) 200 (D) 50

Ans. B

Sol. $$\sum_{i=1}^{20}\left(\frac{{}^{20}C_{i-1}}{{}^{20}C_i + {}^{20}C_{i-1}}\right)^3 = \frac{k}{21}$$

$$\sum_{i=1}^{20}\left(\frac{{}^{20}C_{i-1}}{{}^{21}C_i}\right)^3 = \frac{k}{21}$$

$$\sum_{i=1}^{20}\left(\frac{i}{21}\right)^3 = \frac{k}{21}$$

$$\frac{1}{(21)^3}\left(\frac{20\times 21}{2}\right)^2 = \frac{k}{21}$$

$k = 100$

257. Let $(x + 10)^{50} + (x - 10)^{50} = a_0 + a_1x + a_2x^2 + ...+ a_{50}x^{50}$, for all $x \in R$; then $\frac{a_2}{a_0}$ is equal to :

(A) 12.75 (B) 12.25 (C) 12.50 (D) 12.00

माना सभी $x \in R$ के लिए $(x + 10)^{50} + (x - 10)^{50} = a_0 + a_1x + a_2x^2 + ...+ a_{50}x^{50}$, तो $\frac{a_2}{a_0}$ बराबर है –

(A) 12.75 (B) 12.25 (C) 12.50 (D) 12.00

Ans. B

Sol. $(x+10)^{50} + (x-10)^{50} = a_0 + a_1x + a_2x^2y$ ..... $a_0 = 2 - 10^{50}$

$50(x + 10)^{49} + 50(x-10)^{49} = a_1 + 2a_2x + ...$

$50 \times 49\,(x + 10)^{48} + 50 \times 49\,(x-10)^{98} = 2a_2 + ...$

$$\frac{50\times 49\times 10^{48} + 50\times 49\times 10^{98}}{2} = a_2$$

$a_2 = 50\times 49\times 10^{48}$

$$\frac{a_2}{a_0} = \frac{50\times 49\times 10^{48}}{2\times 10^{50}} = \frac{50\times 49}{2\times 100} = \frac{49}{4}$$

258. Let S = {1,2,3,.....,100}. The number of non-empty subsets A of S such that the product of elements in A is even is :

(A) $2^{50}(2^{50}-1)$ (B) $2^{50}-1$ (C) $2^{100}-1$ (D) $2^{50}+1$

माना S = {1,2,3,.....,100} तो S के उन सभी अरिक्त उपसमुच्चयों A जिनके अवयवों का गुणनफल सम है, की संख्या है –

(A) $2^{50}(2^{50}-1)$ (B) $2^{50}-1$ (C) $2^{100}-1$ (D) $2^{50}+1$

Ans. A

Sol. Let S = {1,2,3,.....,100}

Total subset = $2^{100}-1$

non empty subset with product of element is even = Total Subsets - (Total Subsets when all elements are odd)

$= 2^{100}-1-(2^{50}-1)$

$= 2^{100}-2^{50}$

$= 2^{50}(2^{50}-1)$

259. The coefficient of t^4 in the expansion of $\left(\frac{1-t^6}{1-t}\right)^3$ is :

(A) 15 (B) 14 (C) 12 (D) 10

$\left(\frac{1-t^6}{1-t}\right)^3$ के प्रसार में t^4 का गुणांक है –

(A) 15 (B) 14 (C) 12 (D) 10

Ans. A

Sol. $(1-t^6)^3(1-t)^{-3}$

$= (1-t^{18}-3t^6+3t^{12})(1-t)^{-3}$

coefficient of t^4 in $(1-t)^{-3}$

$^{3+4-1}C_4 = {}^6C_2 = 15$

260. If some three consecutive coefficients in the binomial expansion of $(x+1)^n$ in powers of x are in the ratio 2:15:70, then the average of these three coefficients is:

(A) 625 (B) 964 (C) 232 (D) 227

यदि $(x+1)^n$ के x की घातों में द्विपद प्रसार में कोई तीन क्रमागत गुणांक 2:15:70 के अनुपात में हैं, तो इन तीन गुणांकों का औसत है :

(A) 625 (B) 964 (C) 232 (D) 227

Ans. C

Sol. Given $^nC_{r-1} : {}^nC_r : {}^nC_{r+1} = 2 : 15 : 70$

$$\frac{{}^nC_{r-1}}{{}^nC_r} = \frac{2}{15} \;\&\; \frac{{}^nC_r}{{}^nC_{r+1}} = \frac{15}{70}$$

$$\Rightarrow \quad \frac{r}{n-r+1} = \frac{2}{15} \;\&\; \frac{r+1}{n-r} = \frac{3}{14}$$

$\Rightarrow$ $15r = 2n - 2r + 2$ & $14r + 14 = 3n - 3r$

$\Rightarrow$ $17r = 2n + 2$ & $17r = 3n - 14$

i.e., $2n + 2 = 3n - 14 \Rightarrow n = 16$ & $r = 2$

$$\text{Mean} = \frac{{}^{16}C_1 + {}^{16}C_2 + {}^{16}C_3}{3} = \frac{16 + 120 + 560}{3}$$

$$= \frac{696}{3} = 232$$

261. If the coefficients of x^2 and x^3 are both zero, in the expansion of the expression $(1 + ax + bx^2)(1 - 3x)^{15}$ in powers of x, then the ordered pair (a, b) is equal to :

(A) (– 54, 315) (B) (28, 861) (C) (–21, 714) (D) (28, 315)

यदि x की घातों (powers) में, व्यंजक $(1 + ax + bx^2)(1 - 3x)^{15}$ के प्रसार में x^2 तथा x^3 दोनों के गुणांक शून्य के बराबर हैं, तो क्रमित युग्म (a, b) बराबर है :

(A) (– 54, 315) (B) (28, 861) (C) (–21, 714) (D) (28, 315)

Ans. D

Sol. $(1 + ax + bx^2)(1 - 3x)^{15}$

Co-eff. of $x^2 = 1.{}^{15}C_2(-3)^2 + a.{}^{15}C_1(-3) + b.{}^{15}C_0$

$$= \frac{15 \times 14}{2} \times 9 - 15 \times 3a + b = 0 \text{ (Given)}$$

$\Rightarrow$ $945 - 45a + b = 0$ ...(i)

Now co-eff. of $x^3 = 0$

$\Rightarrow$ ${}^{15}C_3(-3)^3 + a.{}^{15}C_2(-3)^2 + b.{}^{15}C_1(-3) = 0$

$$\Rightarrow \frac{15 \times 14 \times 13}{3 \times 2} \times (-3 \times 3 \times 3) + a \times \frac{15 \times 14 \times 9}{2} - b \times 3 \times 15 = 0$$

$\Rightarrow$ $15 \times 3[-3 \times 7 \times 13 + a \times 7 \times 3 - b] = 0$

$\Rightarrow$ $21a - b = 273$ ...(ii)

From (i) and (ii)

$a = +28, b = 315 \equiv (a, b) \equiv (28, 315)$

262. The smallest natural number n, such that the coefficient of x in the expansion of $\left(x^2 + \frac{1}{x^3}\right)^n$ is ${}^nC_{23}$, is :

(A) 58 (B) 35 (C) 38 (D) 23

वह न्यूनतम प्राकृत संख्या n, जिसके लिए $\left(x^2 + \frac{1}{x^3}\right)^n$ के प्रसार में x का गुणांक ${}^nC_{23}$ है, है :

(A) 58 (B) 35 (C) 38 (D) 23

Ans. C

Sol. $\left(x^2+\frac{1}{x^3}\right)^n$

General term $T_{r+1} = {}^nC_r(x^2)^{n-r}\left(\frac{1}{x^3}\right)^r$

${}^nC_r \cdot x^{2n-5r}$

for coefficiant of x, $2n-5r=1$

Given ${}^nC_r = {}^nC_{23}$

$r = 23$ or $n-r=23$

$\Rightarrow n = 58$ or $n = 38$

Minimum value is $n = 38$

263. The coefficient of x^{18} in the product $(1+x)(1-x)^{10}(1+x+x^2)^9$ is :

(A) 84 (B) –126 (C) –84 (D) 126

गुणनफल $(1+x)(1-x)^{10}(1+x+x^2)^9$ में x^{18} का गुणांक है :

(A) 84 (B) –126 (C) –84 (D) 126

Ans. A

Sol. $(1-x)^{10}(1+x+x^2)^9(1\times x)$

$= (1-x^3)^9(1-x^2)$

$= (1-x^3)^9 - x^2(1-x^3)^9$

$\Rightarrow$ Coefficient of x^{18} in $(1-x^3)^9$ – coeff. of x^{16} in $(1-x^3)^9$

${}^9C_6 = \frac{9!}{6!3!} = \frac{7\times8\times9}{6} = 84$

264. If ${}^{20}C_1 + (2^2)\,{}^{20}C_2 + (3^2)\,{}^{20}C_3 + \ldots\ldots\ldots + (20^2)\,{}^{20}C_{20} = A(2^\beta)$, then the ordered pair (A, β) is equal to

(A) (420,19) (B) (380,19) (C) (420,18) (D) (380,18)

यदि ${}^{20}C_1 + (2^2)\,{}^{20}C_2 + (3^2)\,{}^{20}C_3 + \ldots\ldots\ldots + (20^2)\,{}^{20}C_{20} = A(2^\beta)$, तो क्रमित युग्म (A, β) बराबर है :

(A) (420,19) (B) (380,19) (C) (420,18) (D) (380,18)

Ans. C

Sol. ${}^{20}C_1 + 2^2 \cdot {}^{20}C_2 + 3^2 \cdot {}^{20}C_3 + \ldots + 20^2 \cdot {}^{20}C_{20}$

$= \sum_{r=1}^{20} r^2 \cdot {}^{20}C_r$

$= 20\sum_{r=1}^{20} r \cdot {}^{19}C_{r-1}$

$$= 20\left[\sum_{r=1}^{20}(r-1)^{19}C_{r-1} + \sum_{r=1}^{20}{}^{19}C_{r-1}\right]$$

$$= 20\left[19\sum_{r=2}^{20}{}^{18}C_{r-2} + 2^{19}\right]$$

$= 20[19 \cdot 2^{18} + 2^{19}]$

$= 20 \times 21 \times 2^{18}$

$= 420 \times 2^{18}$

So; $A = 420$ and $\beta = 18$

265. The term independent of x in expansion of $\left(\frac{1}{60} - \frac{x^8}{81}\right)\cdot\left(2x^2 - \frac{3}{x^2}\right)^6$ is equal to

(A) –108 (B) –36 (C) –72 (D) 36

$\left(\frac{1}{60} - \frac{x^8}{81}\right)\cdot\left(2x^2 - \frac{3}{x^2}\right)^6$ के प्रसार में x से स्वतंत्र पद है :

(A) –108 (B) –36 (C) –72 (D) 36

Ans. B

Sol. $\left(\frac{1}{60} - \frac{x^8}{81}\right)\left(2x^2 - \frac{3}{x^2}\right)^6$

$$= \frac{1}{60}\left(2x^2 - \frac{3}{x^2}\right)^6 - \frac{x^8}{81}\left(2x^2 - \frac{3}{x^2}\right)^6$$

Coefficient of x^o in $= \frac{1}{60}\left(2x^2 - \frac{3}{x^2}\right)^6 - \frac{x^8}{81}$

coefficient of x^{-8} in $\left(2x^2 - \frac{3}{x^2}\right)^6$

$$= \frac{-1}{60}{}^6C_3(2)^3(3)^3 + \frac{1}{81}{}^6C_5(2)(3)^5$$

$= -72 + 36$

$= -36$

266. The greatest positive integer k, for which $49^k + 1$ is a factor of the sum $49^{125} + 49^{124} + \ldots\ldots 49^2 + 49 + 1$, is :

(A) 60 (B) 63 (C) 65 (D) 32

सबसे बड़ी धन पूर्णांक संख्या k, जिसके लिए $49^k + 1$ योगफल $49^{125} + 49^{124} + \ldots\ldots 49^2 + 49 + 1$ का एक गुणन खंड है, है :

(A) 60 (B) 63 (C) 65 (D) 32

Ans. B

Sol. $\text{Sum} = \frac{49^{126}-1}{49-1} = \frac{\left(49^{63}+1\right)\left(49^{63}-1\right)}{49-1}$

So k = 63

267. The number of ordered pairs (r, k) for which $6.^{35}C_r = (k^2-3).^{36}C_{r+1}$, where k is an integer, is :

(A) 3 (B) 4 (C) 2 (D) 6

क्रमित युग्मों (r, k), जिनके लिए $6.^{35}C_r = (k^2-3).^{36}C_{r+1}$, जहाँ k एक पूर्णांक है, की संख्या है :

(A) 3 (B) 4 (C) 2 (D) 6

Ans. B

Sol. $(6)\ (^{35}C_r) = k^2 - 3\ (^{36}C_{r+1})$

$$\frac{k^2-3}{6} = \frac{^{35}C_r}{^{36}C_{r+1}}$$

$$\frac{k^2-3}{6} = \frac{r+1}{36}$$

$6k^2 = r + 19$

$k = 2 \Rightarrow r = \pm 5$ $\quad$ $k = 3 \Rightarrow r = 35$

$k = -2 \Rightarrow r = 5$ $\quad$ $k = -3 \Rightarrow r = 35$

Total no. of ordered pairs (r, k) is 4

268. If α and β be the coefficients of x^4 and x^2 respectively in the expansion of $\left(x+\sqrt{x^2-1}\right)^6 + \left(x-\sqrt{x^2-1}\right)^6$, then:

(A) $\alpha + \beta = -30$ (B) $\alpha + \beta = 60$ (C) $\alpha - \beta = 60$ (D) $\alpha - \beta = -132$

यदि $\left(x+\sqrt{x^2-1}\right)^6 + \left(x-\sqrt{x^2-1}\right)^6$ के प्रसार में x^4 तथा x^2 के गुणांक क्रमशः α तथा β हैं, तो :

(A) $\alpha + \beta = -30$ (B) $\alpha + \beta = 60$ (C) $\alpha - \beta = 60$ (D) $\alpha - \beta = -132$

Ans. D

Sol. $\left(x+\sqrt{x^2-1}\right)^6 + \left(x-\sqrt{x^2-1}\right)^6 = 2\left[{}^6C_0\, x^6 + {}^6C_2 x^4(x^2-1) + {}^6C_4\, x^2(x^2-1) + {}^6C_6(x^2-1)^3\right]$

$= 2\left[32x^6 - 48x^4 + 18x^2 - 1\right]$

Coefficient of $x^2(\alpha) = -96$

Coefficient of $x^2(\beta) = 36$ $\quad$ $\alpha - \beta = -132$

269. In the expansion of $\left(\frac{x}{\cos\theta} + \frac{1}{x\sin\theta}\right)^{16}$, if l_1 is the least value of the term independent of x when $\frac{\pi}{8} \le \theta \le \frac{\pi}{4}$ and l_2 is the least value of the term independent of x when $\frac{\pi}{16} \le \theta \le \frac{\pi}{8}$, then the ratio $l_2 : l_1$ is equal to :

(A) 1 : 16 (B) 16 : 1 (C) 1 : 8 (D) 8 : 1

$\left(\frac{x}{\cos\theta}+\frac{1}{x\sin\theta}\right)^{16}$ के प्रसार में, यदि x से स्वतंत्र पद का निम्नतम मान l_1 है जब $\frac{\pi}{8}\le\theta\le\frac{\pi}{4}$ तथा x से स्वतंत्र पद का निम्नतम मान l_2 है जब $\frac{\pi}{16}\le\theta\le\frac{\pi}{8}$, तो अनुपात $l_2 : l_1$ बराबर है :

(A) 1 : 16 (B) 16 : 1 (C) 1 : 8 (D) 8 : 1

Ans. B

Sol. $T_{r+1}={}^{16}C_r\left(\frac{x}{\cos\theta}\right)^{16-r}\left(\frac{1}{x\sin\theta}\right)^{r}$

$= {}^{16}C_r\ x^{16-2r}\times(\cos\theta)^{r-16}(\sin\theta)^{-r}$

$= 16-2r=0 \qquad \Rightarrow r=8$

$= {}^{16}C_8\ (\cos\theta)^{-8}(\sin\theta)^{-8}$

Term independent of $x = {}^{16}C_8\times\frac{1}{(\sin\theta\cos\theta)^8}$

$= {}^{16}C_8\times\frac{2^8}{(\sin 2\theta)^8}$

$l_1 : \frac{\pi}{4}\le 2\theta\le\frac{\pi}{2} \qquad l_2 : \frac{\pi}{8}\le 2\theta\le\frac{\pi}{4}$

$l_1 : {}^{16}C_8\times 2^8 \qquad l_2 : {}^{16}C_8\times 2^{12}$

$l_2 : l_1 \Rightarrow 2^4 : 1 \Rightarrow 16 : 1$

270. The value of

$-{}^{15}C_1+2\cdot{}^{15}C_2-3\cdot{}^{15}C_3+....-15\cdot{}^{15}C_{15}+{}^{14}C_1+{}^{14}C_3+{}^{14}C_5+....+{}^{14}C_{11}$ is :

$-{}^{15}C_1+2\cdot{}^{15}C_2-3\cdot{}^{15}C_3+....-15\cdot{}^{15}C_{15}+{}^{14}C_1+{}^{14}C_3+{}^{14}C_5+....+{}^{14}C_{11}$ का मान है :

(A) $2^{13}-14$ (B) 2^{14} (C) $2^{16}-1$ (D) $2^{13}-13$

Ans. A

Sol. $(1+x)^{15}={}^{15}C_0+{}^{15}C_1x+{}^{15}C_2x^2+....+{}^{15}C_{15}x^{15}$

D w r to x

$15(1+x)^{14}={}^{15}C_1+2\cdot{}^{15}C_2x+....+15\cdot{}^{15}C_{15}x^{14}$

Put $x=-1$

$-{}^{15}C_1+2\cdot{}^{15}C_2+....-15\ {}^{15}C_{15}=0$

${}^{14}C_1+{}^{14}C_3+....+{}^{14}C_{11}+{}^{14}C_{13}=2^{13}$

${}^{14}C_1+{}^{14}C_3+....+{}^{14}C_{11}=2^{13}-14$

$S = 2^{13} - 14$

271. If $n \geq 2$ is a positive integer, then the sum of the series ${}^{n+1}C_2 + 2\left({}^2C_2 + {}^3C_2 + {}^4C_2 + ... + {}^nC_2\right)$ is :

यदि $n \geq 2$ एक धनात्मक पूर्णांक है, तो श्रेणी ${}^{n+1}C_2 + 2\left({}^2C_2 + {}^3C_2 + {}^4C_2 + ... + {}^nC_2\right)$ का योग है :

(A) $\frac{n(n+1)^2(n+2)}{12}$ (B) $\frac{n(n+1)(2n+1)}{6}$

(C) $\frac{n(2n+1)(3n+1)}{6}$ (D) $\frac{n(n-1)(2n+1)}{6}$

Ans. B

Sol. $E = {}^{n+1}C_2 + 2\left({}^3C_3 + {}^3C_2 + {}^4C_2 + ... + {}^nC_2\right)$

$= {}^{n+1}C_2 + 2\,({}^{n+1}C_3)$

$= \frac{(n+1)n}{2} + \frac{2(n+1)n(n-1)}{6}$

$= \frac{n(n+1)(3+2n-2)}{6}$

$= \frac{n(n+1)(2n+1)}{6}$ $\Rightarrow$ option (2)

272. The maximum value of the term independent of 't' in the expansion of $\left(tx^{\frac{1}{5}} + \frac{(1-x)^{\frac{1}{10}}}{t}\right)^{10}$

where $x \in (0, 1)$ is :

$\left(tx^{\frac{1}{5}} + \frac{(1-x)^{\frac{1}{10}}}{t}\right)^{10}$,जहाँ $x \in (0, 1)$ है, के प्रसार में 't' से स्वतंत्र पद का अधिकतम मान है :

(A) $\frac{2.10!}{3\sqrt{3}(5!)^2}$ (B) $\frac{2.10!}{3(5!)^2}$ (C) $\frac{10!}{\sqrt{3}(5!)^2}$ (D) $\frac{10!}{\sqrt{3}(5!)^2}$

Ans. A

Sol. The term independent of t will be middle term :

Middle term $f(x) = {}^{10}C_5 \cdot x \cdot (1-x)^{\frac{1}{2}}$; $x \in (0,1)$.

$f'(x) = {}^{10}C_5 \cdot \left[(1-x)^{\frac{1}{2}} - \frac{x}{2}(1-x)^{-\frac{1}{2}}\right] = 0$

$\Rightarrow 1-x=\frac{x}{2}\Rightarrow\frac{3x}{2}=1\Rightarrow x=\frac{2}{3}.$

Point of maxima.

$$f\left(\frac{2}{3}\right)={}^{10}C_5\times\frac{2}{3}\left(1-\frac{2}{3}\right)^{\frac{1}{2}}$$

$$=\frac{2}{3\sqrt{3}}\times\frac{10!}{5!5!}$$

273. Let [x] denote greatest integer less than or equal to x. If for $n\in N$, then $(1-x+x^3)^n=\sum_{j=0}^{3n}a_jx^j$, then $\sum_{j=0}^{\left[\frac{3n}{2}\right]}a_{2j}+4\sum_{j=0}^{\left[\frac{3n-1}{2}\right]}a_{2j+1}$ is equal to :

माना [x] महत्तम पूर्णांक $\le x$ है। यदि $n\in N$ के लिए $(1-x+x^3)^n=\sum_{j=0}^{3n}a_jx^j$ है, तो $\sum_{j=0}^{\left[\frac{3n}{2}\right]}a_{2j}+4\sum_{j=0}^{\left[\frac{3n-1}{2}\right]}a_{2j+1}$ बराबर है:

(A) n (B) 2^{n-1} (C) 2 (D) 1

Ans. D

Sol. $f(x)=(1-x+x^3)^n$

$f(1)=1, f(-1)=1$

$$\sum a_{2j}=\frac{f(1)+f(-1)}{2}=\frac{1+1}{2}=1$$

$$\sum a_{2j+1}=\frac{f(1)-f(-1)}{2}=\frac{1-1}{2}=0$$

Required sum = 1 + 4 (0) = 1

274. If $({}^{30}C_1)^2+2({}^{30}C_2)^2+3({}^{30}C_3)^2+.....+30({}^{30}C_{30})^2=\frac{\alpha 60!}{(30!)^2}$ then α is equal to :

यदि $({}^{30}C_1)^2+2({}^{30}C_2)^2+3({}^{30}C_3)^2+.....+30({}^{30}C_{30})^2=\frac{\alpha 60!}{(30!)^2}$ है, तो α बराबर है :

(A) 15 (B) 60 (C) 10 (D) 30

Ans. A

Sol. $S=0\cdot\left({}^{30}C_0\right)^2+1\cdot\left({}^{30}C_1\right)^2+2\cdot\left({}^{30}C_2\right)^2+.....+30\cdot\left({}^{30}C_{30}\right)^2$

$$S = 30\cdot\left({}^{30}C_0\right)^2 + 29\cdot\left({}^{30}C_1\right)^2 + 28\cdot\left({}^{30}C_2\right)^2 + \ldots + \cdot 0\cdot\left({}^{30}C_0\right)^2$$

$$2S = 30\cdot\left({}^{30}C_0{}^2 + {}^{30}C_1{}^2 + \ldots\ldots\cdot + {}^{30}C_{30}{}^2\right)$$

$$S = 15\cdot{}^{60}C_{30} = 15\cdot\frac{60!}{(30!)^2}$$

$$\frac{15\cdot 10!}{(30!)^2} = \frac{\alpha\cdot 60!}{(30!)^2}$$

$\Rightarrow \alpha = 15$

275. If a_r is the coefficient of x^{10-r} in the Binomial expansion of $(1+x)^{10}$, then $\sum_{r=1}^{10} r^3\left(\frac{a_r}{a_{r-1}}\right)^2$ is equal to :

यदि $(1+x)^{10}$ के द्विपद प्रसार में x^{10-r} का गुणांक a_r है, तो $\sum_{r=1}^{10} r^3\left(\frac{a_r}{a_{r-1}}\right)^2$ बराबर है :

(A) 4895 (B) 5445 (C) 3025 (D) 1210

Ans. D

Sol. Coeff. of x^{10-r} in $(1+x)^{10}$

$a_r = {}^{10}C_r$

$$\frac{a_r}{a_{r-1}} = \frac{{}^{10}C_r}{{}^{10}C_{r-1}} = \frac{10-r+1}{r} = \frac{11-r}{r}$$

$$\sum_{r=0}^{10} r^3\left(\frac{11-r}{r}\right)^2 = \sum_{r=0}^{10} r(11-r)^2$$

$$= \sum_{r=0}^{10}\left(r^3 - 22r^2 + 121r\right)$$

$$= \sum_{r=0}^{10} r^3 - 22\sum_{r=0}^{10} r^2 + 121\sum_{r=0}^{10} r = \left(\frac{10(10+1)}{2}\right)^2 - 22\times\frac{10(10+1)(20+1)}{6} + 121\times\frac{10\cdot 11}{2}$$

$$= 55^2 - 11\times 11\times 70 + 5\times 11^3 = 11^2[25-70+55] = 11^2\times 10 = 1210$$

276. Let K be the sum of the coefficients of the odd powers of x in the expansion of $(1+x)^{99}$. Let a be the middle term in the expansion of $\left(2+\frac{1}{\sqrt{2}}\right)^{200}$. If $\frac{{}^{200}C_{99}\,K}{a} = \frac{2^{\ell}\,m}{n}$, where m and n are odd numbers, then the ordered pair (l, n) is equal to

माना $(1+x)^{99}$ के प्रसार में x की विषम घातों के गुणांको का योग K है। माना $\left(2+\frac{1}{\sqrt{2}}\right)^{200}$ के प्रसार में मध्य पद a है। यदि $\frac{^{200}C_{99}\,K}{a}=\frac{2^{\ell}\,m}{n}$ है, जहाँ *m* तथा *n* विषम संख्याऐं है, तो क्रमित युग्म (*l*, n) बराबर है।

(A) (50, 51) (B) (51,101) (C) (50,101) (D) (51,99)

Ans. C

Sol. Sum of odd coeff. of $x=\frac{2^{99}-0}{2}=k$

$k=2^{98}$

Middle term $\alpha = T101 = {}^{200}C_{100}\frac{(2)^{100}}{2^{50}}$

$= {}^{200}C_{100}\,2^{50}$

Now $\frac{^{200}C^{99}\times k}{\alpha}=\frac{200!}{100!99!}\times\frac{2^{98}}{^{200}C_{100}\times 2^{50}}$

$=\frac{100}{101}\times 2^{48}$

$\frac{25}{101}\times 2^{50}$

277. The coefficient of x^{301} in $(1+x)^{500}+x(1+x)^{499}+x^2(1+x)^{498}+.....+x^{500}$ is :

$(1+x)^{500}+x(1+x)^{499}+x^2(1+x)^{498}+.....+x^{500}$ में x^{301} का गुणांक है :

(A) $^{501}C_{302}$ (B) $^{500}C_{300}$ (C) $^{500}C_{301}$ (D) $^{501}C_{200}$

Ans. D

Sol. $(1+x)^{500}+x(1+x)^{499}+x^2(1+x)^{498}+....+x^{500}$

$=(1+x)^{500}\left\{\frac{1-\left(\frac{x}{1+x}\right)^{501}}{1-\frac{x}{1+x}}\right\}$

$=(1+x)^{500}\frac{\left((1+x)^{501}-x^{501}\right)}{(1+x)^{501}}\cdot(1+x)$

$=(1+x)^{501}-x^{501}$

Coefficient of x^{301} in $(1+x)^{501}-x^{501}$ is given by $^{501}C_{301}=\,^{501}C_{200}$

278. Let $x=\left(8\sqrt{3}+13\right)^{13}$ and $y=\left(7\sqrt{2}+9\right)^{9}$. If [t] denotes the greatest integer $\le t$, then

(A) [x] is even but [y] is odd (B) [x] + [y] is even

(C) [x] and [y] are both odd (D) [x] is odd but [y] is even

माना $x=\left(8\sqrt{3}+13\right)^{13}$ और $y=\left(7\sqrt{2}+9\right)^{9}$ है। यदि [t] महत्तम पूर्णांक $\le t$ है, तब

(A) [x] सम है परन्तु [y] विषम है (B) [x] + [y] सम है

(C) [x] एवं [y] दोनों विषम है (D) [x] विषम है परन्तु [y] सम है

Ans. B

Sol. $x=\left(8\sqrt{3}+13\right)={}^{13}C_0\cdot\left(8\sqrt{3}\right)^{13}+{}^{13}C_1\left(8\sqrt{3}\right)^{12}+....$

$x'=\left(8\sqrt{3}-13\right)^{13}={}^{13}C_0\left(8\sqrt{3}\right)^{13}-{}^{13}C_1\left(8\sqrt{3}\right)^{12}(13)^1+...$

$x-x'=2\left[{}^{13}C_1\cdot\left(8\sqrt{3}\right)^{12}(13)^1+{}^{13}C_3\left(8\sqrt{3}\right)^{10}\cdot(13)^3....\right]$

therefore, $x-x'$ is even integer, hence [x] is evem

Now, $y=\left(7\sqrt{2}+9\right)^{9}={}^{9}C_0\left(7\sqrt{2}\right)^{9}+{}^{9}C_1\left(7\sqrt{2}\right)^{8}(9)^1+{}^{9}C_2\left(7\sqrt{2}\right)^{7}(9)^2.....$

$y'=\left(7\sqrt{2}-9\right)^{9}={}^{9}C_0\left(7\sqrt{2}\right)^{9}-{}^{9}C_1\left(7\sqrt{2}\right)^{8}(9)^1+{}^{9}C_2\left(7\sqrt{2}\right)^{7}(9)^2.....$

$y-y'=2\left[{}^{9}C_1\left(7\sqrt{2}\right)^{8}(9)^1+{}^{9}C_3\left(7\sqrt{2}\right)^{6}(9)^3+...\right]$

$y-y'=$ Even integer, hence [y] is even

279. The value of $\frac{1}{1!50!}+\frac{1}{3!48!}+\frac{1}{5!46!}+.....+\frac{1}{49!2!}+\frac{1}{51!1!}$ is :

$\frac{1}{1!50!}+\frac{1}{3!48!}+\frac{1}{5!46!}+.....+\frac{1}{49!2!}+\frac{1}{51!1!}$ का मान है :

(A) $\frac{2^{51}}{51!}$ (B) $\frac{2^{50}}{51!}$ (C) $\frac{2^{51}}{50!}$ (D) $\frac{2^{50}}{50!}$

Ans. B

Sol. $\sum_{r=1}^{26}\frac{1}{(2r-1)!(51-(2r-1))!}=\sum_{r=1}^{26}{}^{51}C_{(2r-1)}\frac{1}{51!}$

$=\frac{1}{51!}\left\{{}^{51}C_1+{}^{51}C_3+...+{}^{51}C_{51}\right\}=\frac{1}{51!}\left(2^{50}\right)$

280. The sum $\sum_{n=1}^{\infty}\frac{2n^2+3n+4}{(2n)!}$ is equal to :

योगफल $\sum_{n=1}^{\infty}\frac{2n^2+3n+4}{(2n)!}$ बराबर है :

(A) $\frac{13e}{4}+\frac{5}{4e}$ (B) $\frac{13e}{4}+\frac{5}{4e}-4$

(C) $\frac{11e}{2}+\frac{7}{2e}$ (D) $\frac{11e}{2}+\frac{7}{2e}-4$

Ans. B

Sol. $\sum_{n=1}^{\infty}\frac{2n^2+3n+4}{2n!}$

$=\frac{1}{2}\sum_{n=1}^{\infty}\frac{2n(2n-1)+8n+8}{2n!}$

$=\frac{1}{2}\sum_{n=1}^{\infty}\left\{\frac{1}{(2n-2)!}+\frac{4}{(2n-1)!}+\frac{8}{2n!}\right\}$

$=\frac{1}{2}\left\{\left(\frac{e+e^{-1}}{2}\right)+4\left(\frac{e-e^{-1}}{2}\right)+8\left(\frac{e+e^{-1}}{2}-1\right)\right\}$

$=\frac{1}{4}\left(13e+5e^{-1}-16\right)$

$=\frac{13}{4}e+\frac{5}{4e}-4$

281. If the ratio of the fifth term from the beginning to the fifth term from the end in the expansion of $\left(\sqrt[4]{2}+\frac{1}{\sqrt[4]{3}}\right)^n$ is $\sqrt{6}:1$, then the third term from the beginning is :

यदि $\left(\sqrt[4]{2}+\frac{1}{\sqrt[4]{3}}\right)^n$ के विस्तार में आरंभ से पाँचवें पद का अंत से पाँचवे पद से अनुपात $\sqrt{6}:1$ है, तब आरंभ से तीसरा पद है :

(A) $30\sqrt{3}$ (B) $60\sqrt{3}$ (C) $60\sqrt{2}$ (D) $30\sqrt{2}$

Ans. B

Sol. $$\frac{{}^{n}C_4 2^{\frac{n-4}{4}} \cdot \left(3^{\frac{-1}{4}}\right)^4}{{}^{n}C_4 3^{-\left(\frac{n-4}{4}\right)} \cdot \left(2^{\frac{1}{4}}\right)^4} = \frac{\sqrt{6}}{1}$$

$\Rightarrow n = 10$

So $T_3 = {}^{10}C_2 2^{\frac{1}{4}\cdot 8} \cdot 3^{-\frac{1}{4}\cdot 2} = \frac{45.4}{\sqrt{3}} = 60\sqrt{3}$

282. If the coefficients of x^7 in $\left(ax^2 + \frac{1}{2bx}\right)^{11}$ and x^{-7} in $\left(ax - \frac{1}{3bx^2}\right)^{11}$ are equal, then :

यदि $\left(ax^2 + \frac{1}{2bx}\right)^{11}$ में x^7 तथा $\left(ax - \frac{1}{3bx^2}\right)^{11}$ में x^{-7} के गुणांक बराबर है, तो :

(A) $64ab = 243$ (B) $243ab = 64$ (C) $729ab = 32$ (D) $32ab = 729$

Ans. C

Sol. Coefficient of x^7 in $\left(ax^2 + \frac{1}{2bx}\right)^{11}$

$T_{r+1} = {}^{11}C_r \left(ax^2\right)^{11-r} \left(\frac{1}{2bx}\right)^r$

$= {}^{11}C_r (a)^{11-r} \left(\frac{1}{2b}\right)^r x^{22-3r}$

$22 - 3r = 7 \Rightarrow r = 5$

Coefficient of x^{-7} in $\left(ax - \frac{1}{3bx^2}\right)^{11}$

$T_{r+1} = {}^{11}C_r (ax)^{11-r} \left(-\frac{1}{3bx^2}\right)^r$

$= {}^{11}C_r a^{11-r} \left(-\frac{1}{3b}\right)^r x^{11-3r}$

$\therefore \quad {}^{11-3r} = -7 \Rightarrow r = 6$

$\Rightarrow \quad {}^{11}C_5 (ab)^6 \left(\frac{1}{2b}\right)^5 = 32$

$\Rightarrow \quad 729ab = 32$

283. $25^{190} - 19^{190} - 8^{190} + 2^{190}$ is divisible by :

(A) 14 but not by 34 (B) both 14 and 34 (C) 34 but not by 14 (D) neither 14 nor 34

$25^{190} - 19^{190} - 8^{190} + 2^{190}$

(A) 14 से विभाज्य है परन्तु 34 से नहीं (B) 14 तथा 34 दोनों से विभाज्य है

(C) 34 वे विभाज्य है परन्तु 14 से नहीं (D) न तो 14 से नही 34 से विभाज्य है

Ans. C

Sol. $(25^{190} - 19^{190}) - (8^{190} - 2^{190})$ is divisible by 6.

$(25^{190} - 8^{190}) - (19^{190} - 2^{190})$ is divisible by 17.

$25^{190} - 8^{190}$ is not divisible by 7

but $19^{190} - 2^{190}$ is divisible by 7

So, $25^{190} - 19^{190} - 8^{190} + 2^{190}$ is divisible by 34 but not 14

284. Let the number $(22)^{2022} + (2022)^{22}$ leave the remainder α when divided by 3 and β when divided by 7. Then $(\alpha^2 + \beta^2)$ is equal to :

माना $(22)^{2022} + (2022)^{22}$ को 3 से विभाजित करने पर शेषफल α है तथा 7 से विभाजित करने पर शेषफल β है। तो $(\alpha^2 + \beta^2)$ बराबर है :

(A) 13 (B) 5 (C) 20 (D) 10

Ans. B

Sol. $\text{For } \alpha : (21+1)^{2022} + \underbrace{(2022)^{22}}_{\text{divisible by 3}}$

$= 3K_1 + 1$

$\text{For } \beta : (21+1)^{2022} + (2023-1)^{22}$

$= 7\lambda + 1 + 7\mu + 1$

$= 7K_2 + 2$

So, $\alpha = 1, \beta = 2$

$\alpha^2 + \beta^2 = 5$

285. If the 1011th term from the end in the binomial expansion of $\left(\frac{4x}{5} - \frac{5}{2x}\right)^{2022}$ is 1024 times 1011th term from the beginning, then $|x|$ is equal to :

यदि $\left(\frac{4x}{5} - \frac{5}{2x}\right)^{2022}$ के द्विपद प्रसार में अंत से 1011वाँ पद, आरंभ से 1011वें पद का 1024 गुना है, तो $|x|$ बराबर है :

(A) 8 (B) 12 (C) 10 (D) 15

Ans. C

Sol. T_{1011} from beginning $= T_{1010+1}$

$$= {}^{2022}C_{1010}\left(\frac{4x}{5}\right)^{1012}\left(\frac{-5}{2x}\right)^{1010}$$

T_{1011} from end

$$= {}^{2022}C_{1010}\left(\frac{-5}{2x}\right)^{1012}\left(\frac{4x}{5}\right)^{1010}$$

$$\text{Given}: = {}^{2022}C_{1010}\left(\frac{-5}{2x}\right)^{1012}\left(\frac{4x}{5}\right)^{1010}$$

$$= 2^{10}\cdot{}^{2022}C_{1010}\left(\frac{-5}{2x}\right)^{1010}\left(\frac{4x}{5}\right)^{1012}$$

$$\left(\frac{-5}{2x}\right)^{2} = 2^{10}\left(\frac{4x}{5}\right)^{2}$$

$$x^4 = \frac{5^4}{2^{16}}$$

$$|x| = \frac{5}{16}$$

286. The sum, of the coefficients of the first 50 terms in the binomial expansion of $(1-x)^{100}$, is equal to :

$(1-x)^{100}$ के द्विपद प्रसार में प्रथम 50 पदों के गुणांकों का योग बराबर है :

(A) ${}^{99}C_{49}$ (B) $-{}^{99}C_{49}$ (C) $-{}^{101}C_{50}$ (D) ${}^{101}C_{50}$

Ans. B

Sol. $(1-x)^{100} = Co - C_1x + C_2x^2 - C_3x^3 + \ldots C_{99}x^{99} + C_{100}x^{100}$

$$\Rightarrow Co - C_1 + C_2 - C_3 + \ldots\ldots - C_{99} + C_{100} = 0$$

$$2\left(Co - C_1 + C_2 + \ldots\ldots - C_9\right) + C_{50} = 0$$

$$C_0 - C_1 + C_2 + \ldots C_{99} = -\frac{1}{2}{}^{100}C_{50}$$

$$-\frac{1}{2}\frac{100!}{50!50!} = -\frac{1}{2}\times\frac{100\times 99!}{50!50!} = -{}^{99}C_{49}$$

287. Let $\left(a+bx+cx^2\right)^{10} = \sum_{i=0}^{20} p_i x^i, a,b,c \in \mathbb{N}$. If $p_1 = 20$ and $p_2 = 210$, then 2(a+b+c) is equal to :

यदि $\left(a+bx+cx^2\right)^{10} = \sum_{i=0}^{20} p_i x^i, a,b,c \in \mathbb{N}$, $p_1 = 20$ तथा $p_2 = 210$ हो, तो 2(a+b+c) का मान बराबर होगा :

(A) 8 (B) 12 (C) 6 (D) 15

Ans. B

Sol. $\left(a+bx+cx^2\right)^{10}=\sum_{i=0}^{20}p_i x^i,$

Coefficient of $x^1=20$

$20=\frac{10!}{9!1!}\times a^9\times b^1$

$a^9\cdot b=2$

$a=1, b=2$

Coefficient of $x^2=210$

$210=\frac{10!}{9!1!}\times a^9\times c^1+\frac{10!}{8!2!}\times a^8b^2$

$210=10\cdot c+45\times 4$

$10c=30$

$c=3$

$2(a+b+c)=12$

288. If $\sum_{r=0}^{10}\left(\frac{10^{r+1}-1}{10^r}\right)\cdot{}^{11}C_{r+1}=\frac{\alpha^{11}-11^{11}}{10^{10}}$, then α is equal to :

यदि $\sum_{r=0}^{10}\left(\frac{10^{r+1}-1}{10^r}\right)\cdot{}^{11}C_{r+1}=\frac{\alpha^{11}-11^{11}}{10^{10}}$ है, तो α बराबर है :

(A) 15 (B) 20 (C) 24 (D) 11

Ans. B

Sol. $\sum_{r=0}^{10}\left(\frac{10^{r+1}-1}{10^r}\right)\cdot{}^{11}C_{r+1}$

$=(10-1)\ {}^{11}C_1+\left(\frac{10^2-1}{10}\right)\ {}^{11}C_2+.....+\left(\frac{10^{11}-1}{10^{10}}\right)\ {}^{11}C_{11}$

$=10({}^{11}C_1+{}^{11}C_2+.....+{}^{11}C_{11})-\left({}^{11}C_1+\frac{{}^{11}C_2}{10}+....+\frac{{}^{11}C_{11}}{10^{10}}\right)$

$=10\left(2^{11}-1\right)-10\left(\frac{{}^{11}C_1}{10}+\frac{{}^{11}C_2}{10^2}+....+\frac{{}^{11}C_{11}}{10^{11}}\right)$

$=10\left(2^{11}-1\right)-10\left(\left(1+\frac{1}{10}\right)^{11}-1\right)$

$$=10\left(2^{11}-1\right)-10\left(\frac{11^{11}-10^{11}}{10^{11}}\right)$$

$$=10\left(2^{11}-1\right)-\left(\frac{11^{11}-10^{11}}{10^{10}}\right)$$

$$=\frac{10^{11}\left(2^{11}-1\right)-11^{11}+10^{11}}{10^{10}}$$

$$=\frac{2^{11}\left(10^{11}\right)-11^{11}}{10^{10}}=\frac{20^{11}-11^{11}}{10^{10}}$$

$\Rightarrow \alpha = 20$

289. If the sum of the coefficients of all even powers of x in the product $(1+x+x^2+.....x^{2n})(1-x+x^2-x^3....+x^{2n})$ is 61, then n is equal to _____.

यदि गुणनफल $(1+x+x^2+.....x^{2n})(1-x+x^2-x^3....+x^{2n})$ में x के सभी सम–घातों वाले गुणांकों का यांगफल 61 है, तो n बराबर है _____।

Ans. 30

Sol. Let $(1+x+x^2+.....x^{2n})(1-x+x^2-x^3+..... x^{2n})$

$= a_0 + a_1 x + a_2x^2 + a_3x^3 + ...a_{4n}x^{4n}$ .....(1)

Put x = 1 in equation (1)

$2n + 1 = a_0 + a_1 + a_2 + a_3 + ...a_{4n}$ .....(2)

Put x = –1 in equation (1)

$2n + 1 = a_0 - a_1 + a_2 - a_3 + ...a_{4n}$ .....(3)

Adding (2) and (3)

$2(a_0 + a_2 + a_4 + ...a_{4n}) = 4n + 2$

$\Rightarrow 2 \times 61 = 2(2n + 1)$ $\Rightarrow 2n + 1 = 61$ $\Rightarrow n = 30$

290. Let $\left(2x^2+3x+4\right)^{10}=\sum_{r=0}^{20}a_r x^r$. Then $\frac{a_7}{a_{13}}$ is equal to _____.

माना $\left(2x^2+3x+4\right)^{10}=\sum_{r=0}^{20}a_r x^r$, तो $\frac{a_7}{a_{13}}$ बराबर है _____।

Ans. 8

Sol. $(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r$

General term of $(2x^2 + 3x + 4)^{10}$

$$= \frac{10!}{p_1!p_2!p_3!}\left(2x^2\right)^{p_1}(3x)^{p_2}(4)^{p_3}$$

Where $p_1 + p_2 + p_3 = 10$

and $0 \le p_i \le 10; p_i \in W, i = 1,2,3$

$a_7 =$ coefficient of x^7

$a_{13} =$ coefficient of x^{13}.

$$= \frac{10!}{p_1!p_2!p_3!}2^{p_1}.3^{p_2}.4^{p_3}x^{2p_1+p_2}$$

For a_7 : $2p_1 + p_2 = 7$

p_1	p_2	p_3	Coefficient of x^7
3	1	6	$\frac{10!}{3!.1!.6!}2^3.3^1.4^6 = \frac{10!}{3!.1!.6!}2^{15}.3^1$
2	3	5	$\frac{10!}{2!.3!.5!}2^2.3^3.4^5 = \frac{10!}{2!.3!.5!}2^{12}.3^3$
1	5	4	$\frac{10!}{1!.5!.4!}2^1.3^5.4^4 = \frac{10!}{1!.5!.4!}2^9.3^5$
0	7	3	$\frac{10!}{7!.3!}1.3^7.4^3 = \frac{10!}{7!.3!}2^6.3^7$

$$a^7 = 2^3\left(\frac{10!}{3!.1!.6!}2^{12}.3^1 + \frac{10!}{2!.3!.5!}2^9.3^3 + \frac{10!}{1!.5!.4!}2^6.3^5 + \frac{10!}{7!.3!}2^3.3^7\right)$$

For a_3 : $2p_1 + p_2 = 13$

p_1	p_2	p_3	Coefficient of x^{13}
6	1	3	$\frac{10!}{6!.1!.3!}2^6.3^1.4^3 = \frac{10!}{6!.1!.3!}2^{12}.3^1$
5	3	2	$\frac{10!}{5!.3!.2!}2^5.3^3.4^2 = \frac{10!}{5!.3!.2!}2^9.3^3$
4	5	1	$\frac{10!}{4!.5!.1!}2^4.3^5.4^1 = \frac{10!}{4!.5!.1!}2^6.3^5$
3	7	0	$\frac{10!}{3!.7!}2^3.3^7 = \frac{10!}{3!.7!}2^3.3^7$

$$a^{13} = \left(\frac{10!}{6!.1!.3!}2^{12}.3^1 + \frac{10!}{5!.3!.2!}2^9.3^3 + \frac{10!}{4!.5!.1!}2^6.3^5 + \frac{10!}{3!.7!}2^3.3^7\right)$$

$$\frac{a_7}{a_{13}} = 2^3 = 8$$

291. The natural number m, for which the coefficient of x in the binomial expansion of $\left(x^m+\frac{1}{x^2}\right)^{22}$ is 1540, is ____.

प्राकृत संख्या m, जिसके लिए $\left(x^m+\frac{1}{x^2}\right)^{22}$ के द्विपद प्रसार में x का गुणांक 1540 है,है is ____।

Ans. 13

Sol. General term $T_{r+1}={}^{22}C_r\left(x^m\right)^{22-r}.\left(\frac{1}{x^2}\right)^r$

$={}^{22}C_r\,x^{22m-mr-2r}$

$22m-mr-2r=1$ ..... (1)

and ${}^{22}C_r=1540$

${}^{22}C_3=\frac{22\times21\times20}{3\times2\times1}=1540$

So r = 3 or 19.

If r = 3, $22m-3m-6=1$

$\Rightarrow m=\frac{7}{19}$ (not an integer)

If r = 19, $22m-19m-38=1$

$\Rightarrow m=13$

292. If the remainder when x is divided by 4 is 3, then the remainder when $(2020+x)^{2022}$ is divided by 8 is ____ .

यदि x को 4 से विभाजित करने पर शेषफल 3 है, तो $(2020+x)^{2022}$ को 8 से विभाजित करने पर शेषफल है _____ ।

Ans. 1

Sol. $x=4k+3$

$(2020+x)^{2022}=(2023+4k)^{2022}$

$(2024+(4k-1))^{2022}$

$={}^{2022}C_0(2024)^{2022}+.....+{}^{2022}C_{2021}(2024)(4k-1)^{2021}+{}^{2022}C_{2022}(4k-1)^{2022}$

$=8I+{}^{2022}C_{2022}(4k-1)^{2022}$

$=8I+(4k-1)^{2022}$

$=8I+{}^{2022}C_0(4k)^{2022}+.....+{}^{2022}C_{2022}(-1)^{2022}$

$=8I+8I_1+1$

Remainder = 1

293. If $(2021)^{3762}$ is divided by 17, then the remainder is _____.

यदि $(2021)^{3762}$ को 17 से विभाजित करने पर शेषफल _____।

Ans. 4

Sol. $(2021)^{3768} = (2023\text{-}z)^{3762}$

$= 17\,k + 2^{3762} \quad \{k \in I\}$

$= 17\,k + 2^2\,(2^4)^{940}$

$= 17\,k + 4\,(17-1)^{940}$

$= 17\,k + 4\,(17\lambda + 1) \quad \{\lambda \in I\}$

$= 17k + 4\,(17\lambda) + 4$

Remainder = 4

294. Let n be a positive integer . Let $A = \sum_{k=0}^{n} (-1)^k {}^nC_k \left[\left(\frac{1}{2}\right)^k + \left(\frac{3}{4}\right)^k + \left(\frac{7}{8}\right)^k + \left(\frac{15}{16}\right)^k + \left(\frac{31}{32}\right)^k\right]$. If $63A = 1 - \frac{1}{2^{30}}$,

then n is equal to :

माना n एक धनात्मक पूर्णांक है तथा $A = \sum_{k=0}^{n} (-1)^k {}^nC_k \left[\left(\frac{1}{2}\right)^k + \left(\frac{3}{4}\right)^k + \left(\frac{7}{8}\right)^k + \left(\frac{15}{16}\right)^k + \left(\frac{31}{32}\right)^k\right]$ है। यदि

$63A = 1 - \frac{1}{2^{30}}$ है, तो n बराबर है _____।

Ans. 6

Sol. $A = \sum_{k=0}^{n} (-1)^k {}^nC_k \left[\left(\frac{1}{2}\right)^k + \left(\frac{3}{4}\right)^k + \left(\frac{7}{8}\right)^k + \left(\frac{15}{16}\right)^k + \left(\frac{31}{32}\right)^k\right]$

$A = \left(1 - \frac{1}{2}\right)^n + \left(1 - \frac{3}{4}\right)^n + \left(1 - \frac{7}{8}\right)^n + \left(1 - \frac{15}{16}\right)^n + \left(1 - \frac{31}{32}\right)^n$

$A = \frac{1}{2^n} + \frac{1}{4^n} + \frac{1}{8^n} + \frac{1}{16^n} + \frac{1}{32^n}$

$A = \frac{1}{2^n} + \frac{1}{2^{2n}} + \frac{1}{2^{3n}} + \frac{1}{2^{4n}} + \frac{1}{2^{5n}}$

$$A=\frac{\frac{1}{2^n}\left(\left(\frac{1}{2n}\right)^5-1\right)}{\left(\frac{1}{2n}-1\right)}$$

$$A=\frac{1-2^{5n}}{2^{5n}\left(1-2^n\right)}=\frac{2^{5n}-1}{2^{5n}\left(2^n-1\right)}$$

$$63A=\left(\frac{63}{2^n-1}\right)\left(1-\frac{1}{2^{5n}}\right)=1-\frac{1}{2^{30}}$$

295. If the constant term in the binomial expansion of $\left(\frac{x^{\frac{5}{2}}}{2}-\frac{4}{x^{\ell}}\right)^9$ is -84 and the coefficient of x^{-3l} is $2^{\alpha}\beta$, where $\beta<0$ is an odd number, then $|\alpha\ell-\beta|$ is equal to _____ .

यदि $\left(\frac{x^{\frac{5}{2}}}{2}-\frac{4}{x^{\ell}}\right)^9$ के द्विपद प्रसार में अचर पद -84 है तथा x^{-3l} का गुणांक $2^{\alpha}\beta$ है, जहाँ $\beta<0$ एक विषम संख्या है, तो $|\alpha\ell-\beta|$ बराबर है।

Ans. 98

Sol. In, $\left(\frac{x^{\frac{5}{2}}}{2}-\frac{4}{x^{\ell}}\right)^9$

$$T_{r+1}={}^9C_r\frac{\left(x^{5/2}\right)^{9-r}}{2^{9-r}}\left(\frac{-4}{x^{\ell}}\right)^r$$

$$=(-1)^r\frac{{}^9C_r}{2^{9-r}}4^r x^{\frac{45}{2}-\frac{5r}{2}-lr}$$

$$=45-5r-2lr=0$$

$$r=\frac{45}{5+2l} \qquad \text{_____(1)}$$

Now, according to the question, $(-1)^r\frac{{}^9C_r}{2^{9-r}}4^r=-84$

$= (-1)^r \, {}^9C_r \, 2^{3r-9} = 21 \times 4$

Only natural value of r possible if $3r - 9 = 0$

$r = 3$ and ${}^9C_3 = 84$

$\therefore l = 5$ from equation (1)

Now, coefficient of $x^{-31} = x^{\frac{45}{2} - \frac{5r}{2} - lr}$ at $l = 5$, gives

$r = 5$

$\therefore {}^9C_5 (-1) \frac{4^5}{2^4} = 2^{\alpha} \times \beta$

$= -63 \times 2^7$

$\Rightarrow \alpha = 7, \beta = -63$

$\therefore$ value of $|\alpha l - \beta| = 98$

296. Let the sum of the coefficients of the first three terms in the expansion of $\left(x - \frac{3}{x^2}\right)^n$, $x \neq 0, n \in N$, be 376. Then the coefficients of x^4 is _____.

माना $\left(x - \frac{3}{x^2}\right)^n$, $x \neq 0, n \in N$ के प्रसार में प्रथम तीन पदों के गुणांकों का योग 376 है। तो x^4 का गुणांक _____ है।

Ans. 405

Sol. ${}^nC_0 - {}^nC_1 \cdot 3 + {}^nC_2 \cdot 3^2 = 376$

$\Rightarrow n = 10$

Now $T_{r+1} = {}^{10}C_r \cdot \left(\frac{-3}{x^2}\right)^r \cdot (x)^{10-r}$

$= {}^{10}C_r \cdot (-3)^r \cdot x^{10-3r}$

For $x^4 \Rightarrow 10 - 3r = 4 \Rightarrow r = 2$

Coff. of $x^4 = {}^{10}C_2 \cdot (-3)^2 = 405$

297. The constant term in the expansion of $\left(2x + \frac{1}{x^7} + 3x^2\right)^5$ is _____.

$\left(2x + \frac{1}{x^7} + 3x^2\right)^5$ के प्रसार में अचर पद है।

Ans. 1080

Sol. General term is $\Sigma\frac{5!(2x)^{n_1}\left(x^{-7}\right)^{n_2}\left(3x^2\right)^{n_3}}{n_1!n_2!n_3!}$

For constant term,

$n_1 + 2n_3 = 7n_2$

& $n_1 + n_2 + n_3 = 5$

Only possibility $n_1 = 1, n_2 = 1, n_3 = 3$

$\Rightarrow$ constant term = 1080

298. The remainder when $(2023)^{2023}$ is divided by 35 is _____.

$(2023)^{2023}$ को 35 से विभाजित करने पर शेषफल है।

Ans. 7

Sol. $-72023 = -7 \times 7^{2022}$

$$= \frac{-7^{2022}}{5}$$

$$= \frac{-[50-1]^{1011}}{5} = \frac{[5\lambda-1]}{5}$$

remainder = + 1

when divided by 5 now remainder is 7 when divided 35

299. Let $\sum_{n=0}^{\infty}\frac{n^3((2n)!)+(2n-1)(n!)}{(n!)((2n!))} = ae+\frac{b}{e}+c,$ where $a,b,c \in Z$ and $e=\sum_{n=0}^{\infty}\frac{1}{n!}$. Then $a^2 - b + c$ is equal to

माना $\sum_{n=0}^{\infty}\frac{n^3((2n)!)+(2n-1)(n!)}{(n!)((2n!))} = ae+\frac{b}{e}+c,$ है जहाँ $a,b,c \in Z$ तथा $e=\sum_{n=0}^{\infty}\frac{1}{n!}$ है। तो $a^2 - b + c$ बराबर है।

Ans. 26

Sol. $\sum_{n=0}^{\infty}\frac{n^3(2n)!}{n!(2n)!}+\sum_{n=0}^{\infty}\frac{(2n-1)n!}{n!(2n)!}$

$\sum_{n=0}^{\infty}\frac{(2n-1)n!}{n!(2n)!}$

$\sum_{n=0}^{\infty}\frac{(2n)\times n!}{n!\times(2n)!}-\frac{n!}{n!(2n)!}$

$\sum_{n=0}^{\infty}\frac{1}{(2n-1)!}-\sum_{n=0}^{\infty}\frac{1}{(2n)!}$

$$\left.\frac{e^x - e^{-x}}{2}\right|_{x=1} - \left.\frac{e^x + e^{-x}}{2}\right|_{x=1}$$

$$= \sum_{n=0}^{\infty} \frac{n^3 (2n)!}{n!(2n)!} = \sum_{n=0}^{\infty} \frac{n^3}{n!}$$

$n^3 = an(n-1)(n-2) + bn(n-1) + cn + d$

$n = 1 \Rightarrow c = 1$

$n = 2 \Rightarrow 8 = 2b + 2 \Rightarrow b = 3$

$$= \sum_{n=0}^{\infty} \frac{n(n-1)(n-2) + 3n(n-1) + n}{n!}$$

$$\sum_{n=3}^{\infty} \frac{1}{(n-3)!} + 3\sum_{n=2}^{\infty} \frac{1}{(n-2)!} + \sum_{n=1}^{\infty} \frac{1}{(n-1)!}$$

$$= \sum_{n=0}^{\infty} \frac{n^3 (2n)!}{n!(2n)!} + \sum_{n=0}^{\infty} \frac{(2n-1)n!}{n!(2n)!}$$

$$= 5e + \frac{e}{2} - \frac{1}{e} - \frac{e}{2} + \frac{1}{\frac{e}{2}}$$

$$= 5e - \frac{1}{e}$$

$a = 5, b = -1, c = 0$

Now $a^2 - b + c = 26$

300. 50th root of a number x is 12 and 50th root of another number y is 18. Then the remainder obtained on dividing $(x + y)$ by 25 is _____ .

एक संख्या x का 50वाँ मूल 12 है एवं दूसरी संख्या y का 50वाँ मूल 18 है। तब (x + y) को 25 से विभाजित करने पर प्राप्त शेषफल है।

Ans. 23

Sol. $(x)^{\frac{1}{50}} = 12$ and $(y)^{\frac{1}{50}} = 18$

$\Rightarrow x = 12^{50} \Rightarrow y = 18^{50}$

$x + y = 12^{50} + 18^{50}$

$(2^2 \times 3^1)^{50} + (3^2 \times 2)^{50}$

$= 6^{50} (2^{50} + 3^{50})$

$= (5 + 1)^{50} (4^{25} + 9^{25})$

$= (25\lambda + 1) [(5 - 1)^{25} + (10 - 1)^{25}]$

$= (25\lambda + 1) (25k - 1 + 25\mu - 1)$

$= (25\lambda + 1) (25t + 23)$

$= 25\lambda + 23$

so remiander is 23, when $x + y$ is divided by 25.

301. Let $\alpha > 0$, be the smallest number such that the expansion of $\left(x^{\frac{2}{3}} + \frac{2}{x^3}\right)^{30}$ has term $\beta x^{-\alpha}$, $\beta \in N$. Then α is equal to _____ .

माना $\alpha > 0$ न्यूनतम संख्या है, जिसके लिए $\left(x^{\frac{2}{3}} + \frac{2}{x^3}\right)^{30}$ के प्रसार के एक पद में $\beta x^{-\alpha}$ तो α बराबर है _____ , $\beta \in N$

Ans. 2

Sol. $\alpha > 0, \left(x^{2/3} + \frac{2}{x^3}\right)^{30}$

$T_{r+1} = {}^{30}C_r \left(x^{2/3}\right)^{30-r} \left(\frac{2}{x^3}\right)^r$

$= {}^{30}C_r x^{20-\frac{2r}{3}-3r}$

$= {}^{30}C_r (2)^r x^{\frac{60-11r}{3}} \qquad 0 \le r \le 30$

For $\beta x^{-\alpha}$, $\alpha > 0$ the smallest value of α is 2 (for $r = 6$)

$T7 = {}^{30}C_6 (2)^6 x^{-2}$

$\alpha = 2$

302. The remainder on dividing 5^{99} by 11 is _____ .

5^{99} को 11 से विभाजित करने पर शेषफल _____ है।

Ans. 9

Sol. $5^{99} = 5^4.5^{95}$

$= 625\ [5^5]^{19}$

$= 625\ [3125]^{19}$

$= 625\ [2124 + 1]^{19}$

$= 625\ [11k \times 19 + 1]$

$= 625 \times 11k \times 19 + 625$

$= 11k_1 + 616 + 9$

$= 11(k_2) + 9$

Remainder = 9

303. The remainder, when $19^{200} + 23^{200}$ is divided by 49, is _____.

$19^{200} + 23^{200}$ को 49 से विभाजित करने पर शेषफल _____ है।

Ans. 29

Sol. $(21 - 2)^{200} + (21 + 2)^{200} = 49\lambda + 2^{201}$

$2^{201} = 8^{67} = (7 + 1)^{67} = 49\lambda + 7 \times 67 + 1$

$= 49\lambda + 470$

$= 49 (\lambda + 9) + 29$

So, remainder = 29

304. Let the sixth term in the binomial expansion of $\left(\sqrt{2^{\log_2(10-3^x)}} + \sqrt[5]{2^{(x-2)\log_2 3}}\right)^m$, in the increasing powers of $2^{(x-2)\log_2 3}$, be 21. If the binomial coefficients of the second, third and fourth terms in the expansion are respectively the first, third and fifth terms of an A.P., then the sum of the squares of all possible values of x is _____.

माना $2^{(x-2)\log_2 3}$ की बढ़ती घातों में $\left(\sqrt{2^{\log_2(10-3^x)}} + \sqrt[5]{2^{(x-2)\log_2 3}}\right)^m$ के द्विपद प्रसार में छठा पद 21 है। यदि इस प्रसार में दूसरा, तीसरा तथा चौथा द्विपद गुणांक एक A.P. के क्रमशः पहला, तीसरा तथा पाँचवा पद है, तो x के सभी संभव मानों के वर्गो का योग है।

Ans. 4

Sol. $T_6 = {}^mC_5 \left(10 - 3^x\right)^{\frac{m-5}{2}} \cdot \left(3^{x-2}\right) = 21$ ...(1)

${}^mC_1$, ${}^mC_2$, ${}^mC_3$ are in A.P.

$2.{}^mC_2 = {}^mC_1 + {}^mC_3$

Solving for m, we get

m = 2 (rejected), 7

Put in equation (1)

$21.\left(10 - 3^x\right)\frac{3^x}{9} = 21$

$3^x = 3^0, 3^2$

x = 0, 2

Sum of all the squares of all possible values of x = 4

305. Let α be the constant term in the binomial expansion of $\left(\sqrt{x} - \frac{6}{x^{\frac{3}{2}}}\right)^n$, $n \le 15$. If the sum of the coefficients of the remaining terms in the expansion is 649 and the coefficient of x^{-n} is $\lambda\alpha$, then λ is equal to _____.

माना $\left(\sqrt{x}-\frac{6}{x^{\frac{3}{2}}}\right)^n, n \le 15$ के द्विपद प्रसार में अचर पद α है। यदि इस प्रसार में शेष पदों के गुणांकों का योग 649 है तथा x^{-n} का गुणांक $\lambda\alpha$ है, तो λ बराबर है _____

Ans. 36

Sol. $T_{k+1} = {}^nC_k (x)^{\frac{n-k}{2}} (-6)^k (x)^{\frac{-3}{2}k}$

$\frac{n-k}{2} - \frac{3}{2}k = 0$

$n - 4k = 0$

$(-5)^n - \left({}^nC_{\frac{n}{4}} (-6)^{\frac{n}{4}}\right) = 649$

By observation $(625 + 24 = 649)$, we get $n = 4$

$\because n = 4 \& k = 1$

Required is coefficient of x^{-4} is $\left(\sqrt{4}-\frac{6}{x^{\frac{3}{2}}}\right)^4$

${}^4C_1 (-6)^3$

By calculating we will get $\lambda = 36$

306. The remainder, when 7^{103} is divided by 17, is _____.

7^{103} को 17 से विभाजित करने पर शेषफल है _____

Ans. 12

Sol. 7^{103} divided by 17

$7 \equiv 7 \pmod{17}$

$7^2 \equiv -2 \pmod{17}$

$7^6 \equiv -8 \pmod{17}$

$7^8 \equiv -1 \pmod{17}$

$7^{16} \equiv 1 \pmod{17}$

$7^{103} \equiv 12 \pmod{17}$

$\therefore$ Remainder = 12

307. The number of elements in the set $\{n \in \mathbb{N} : 10 \le n \le 100$ and $3^n - 3$ is a multiple of $7\}$ is _____.

समुच्चय में अवयवों की संख्या होगी $\{n \in \mathbb{N} : 10 \le n \le 100$ तथा $3^n - 3$, 7 का गुणक $\}$_____

Ans. 15

Sol. $n \in [10, 100]$

$3^n - 3$ is multiple of 7

$3^n = 7\lambda + 3$

n = 1,7,13,20,.....97

Number of possible values of n = 15